Measuring drug regulation as a spectrum: Dutch support for nine models of drug regulation

1 Abstract

Rationale Across the globe, drug regulation is being reconsidered. This includes traditionally legal and illegal substances. The Netherlands is no exception. Examples include a national experiment regulating cannabis, tobacco sales being limited to specialty stores and a state commission favourable towards MDMA assisted therapy. These changes are mirrored by changes in public attitudes towards drug regulation.
State of literature Most notably, global support for the regulation of cannabis has risen markedly, while support for the commercial regulation of tobacco (and alcohol?) has declined. In the Netherlands too, public support for fully regulating cannabis has risen to approximately 80%. For most illicit drugs however the picture is less clear. One good source is the 2014 youth Eurobarometer. It found that across EU member states support for regulating mdma hovered between 1% and 15% of Europeans aged 18 to 32.
Literature gap However, the existing literature comes with notable limitations. First, existing measures frame support for drug regulation in binary or ternary terms: prohibition, regulation or unrestricted access. Second, existing samples are not representative of adult national populations. Third, scientific samples are all over a decade old. Fourth, there is limited work modelling attitudes towards drug regulation, in part because it's usually elicited in binary or ternary terms.
Our contribution The present study asks Dutch adults (n=1000) to choose between nine forms of regulated supply for eight different drugs, in decreasing order of restrictiveness: full prohibition, research only, treatment only, on receipt only, sale at speciality store without or with profit, licensed sale by third parties without or with advertisements, and unrestricted access. Further variables collected and an exploratory network analysis modelling support for drug regulation were based on the Causal Attitude Network (CAN) model.
Results

2 Introduction

Shifting regulation worldwide and in the Netherlands Regulatory models which were unthinkable decades ago are now in place: commercial legalization of cannabis in many US states and a ban on tobacco sales to anybody born after 2008 in the UK (Tobacco and Vapes Act, 2026).

Shifting attitudes towards regulation worldwide and in the Netherlands Attitudes towards the regulation of psychoactive substances seem to be changing in the West (Felson et al., 2019; Gallup, 2023).

No proper attitude research trying to understsand this shift As described above, the public is usually consulted in impoverished terms. And government communication is sometimes counterproductive (e.g., messaging that presents drug use as normalised may increase use, van

Wingerde et al., 2023). How we discuss drug regulation matters: in an Irish citizens' assembly 85% rejected the status quo, 58.6% chose a hybrid of different approaches for different drugs, and no universal option drew more than 18.4% (Citizens' Assembly on Drugs Use, 2024).

To date no instrument measures the full structure of beliefs, feelings and behaviour underlying drug regulation preferences (but see Greer & Ritter, 2019; Hamrak & Simonovits, 2026; Wilkins et al., 2022). Similarly, to date no instrument has been developed that measures drug policy preferences as an ordered set of regulatory models per substance. Combining the two yields support for seven regulatory models across nine substances together with the attitude dynamics behind them, usable for instance in public health campaigns.

The scale Drug regulation is inherently multidimensional. For example, more prohibitive policies go hand in hand with more punitive measures (cite?). Nonetheless, we've tried to formulate a unidimensional, and, crucially, a strictly ordinal, scale. Following table ... in MacCoun and Reuter (2001), we define the main dimension of drug regulation as the policy's 'restrictiveness'. This led to the nine options shown in figure 1.

Four other dimensions formed the basis of the attitude beliefs: crime, addiction, the consequences of allowing the sale of drugs, and how drugs fit into society. Belief items were also

For developing our attitude items, we draw on Saris and Gallhofer, 2004's typology of basic concepts-by-intuition, which links the concepts measured in surveys to the linguistic structure of the assertions used to measure them. Evaluations such as "The war on drugs is a failure" have been found to be good predictors of policy stance (Bergquist et al., 2022). However, this may simply reflect their conceptual proximity to the outcome (Vysevkova et al., 2026). Another complicating factor is that people may support a certain causal belief not because they actually believe this but rather as a way to support a norm. For example, support for the effectiveness of punitive measures seems to be driven more by the belief that something must be done, than actually believing that crime will then decrease (cite) (Ajzen & Fishbein, 1977).



Figure 1. The nine answer options.

3 Methods

Sample Two pilots studies were conducted: a convenience sample ($n = 121$) and quota sample of Dutch adults ($n = 120$). Item screening followed the reporting standards discussed in (Burger et al., 2022; Chambon et al., 2026): item redundancy based on ρ, leave-one-out ΔR^2, corrected item-rest. The main study consisted of a single cross-sectional wave taken through Kieskompas (n=1239). The resulting sample was weighted by Kieskompas using iterative post-stratification. Weights were based on the marginal distributions, taken from the Central Bureau of Statistics, of: age, gender, education, migration-background, Nielsen area's, and elicited voting behaviour in last parliamentary elections.

Questionnaire Wording is at B1 Dutch level. Own-use items are anonymous and skippable.

The survey closes with a referral to the Drugs Infolijn.

Analysis. RQ1 is descriptive and exploratory. Per substance the location, dispersion and modality will be reported. The policy sliders are regressed on the attitude items. RQ2 uses a mixed graphical model, estimated with EBIC model selection (Haslbeck & Waldorp, 2020), with bootstrapped edge accuracy and centrality stability (Epskamp et al., 2018). Involvement enters as continuous moderator of edge weights (Haslbeck et al., 2021). Modality is estimated per involvement quartile (Hoffstadt et al., 2026).

3.1 Research questions

.

RQ1. Where does the Dutch public stand on the regulation spectrum, per substance, and what do those positions rest on? Respondents place nine substances on a 0–100 slider anchored by described policy models (see Section **??** for substances and anchors). We describe the distribution per substance and explain it using the attitude items. The result is a map of the debate: which regulatory models are supported, and which beliefs, feelings and behaviours mark out the competing positions.

RQ2. How does the structure of a respondent's attitude network relate to their position on the drug regulation spectrum? The ternary model predicts that cautious, involved respondents hold stable middle positions. Under the binary model an involved respondent sits at one of the extremes. We are interested in under which conditions an involved middle appears.

4 Results

describe figure 2 2

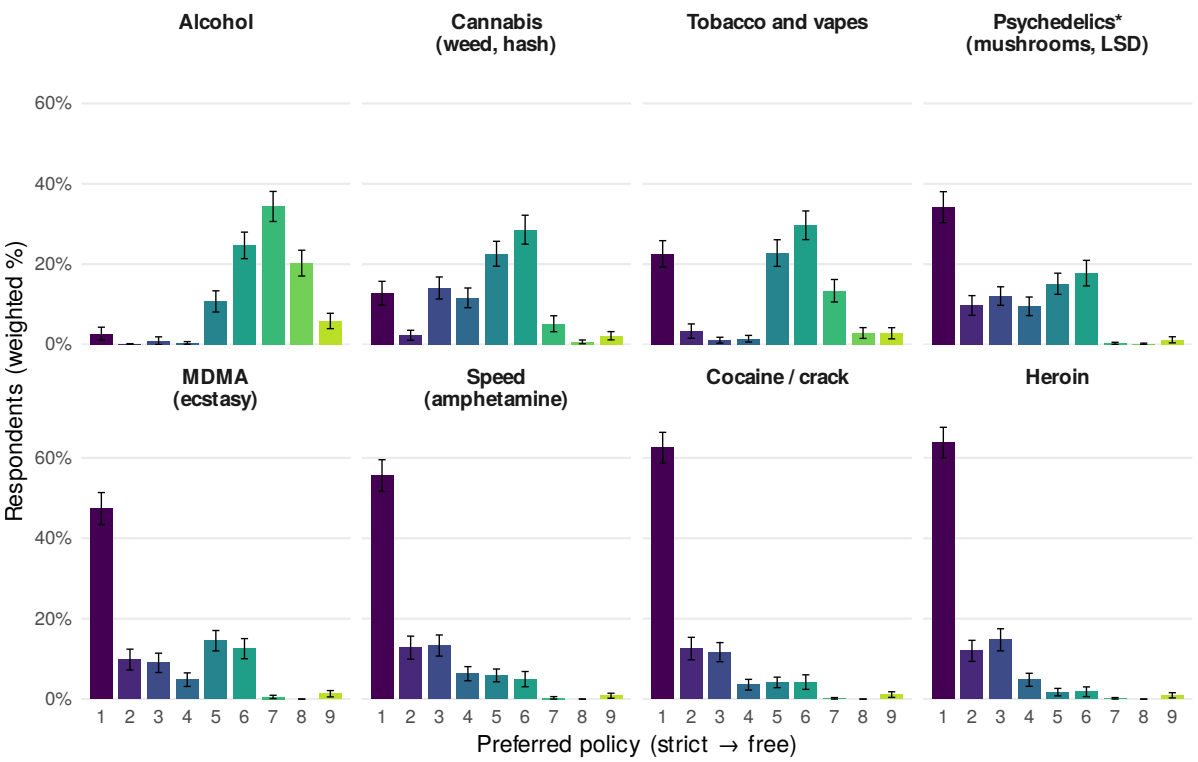


Figure 2. Preferred regulatory model per substance. Question: "Which policy do you think is best?" (*Welk beleid vindt u het beste?*), with nine options from ban (1) to free sale (9); colours and numbers as in Figure 1. Bars show weighted shares with 95% confidence intervals; "no opinion" is excluded ($n = 1225$–1234 per substance). *The item named magic mushrooms and LSD, which are banned in the Netherlands, while psilocybin truffles are legally sold in smartshops; respondents may have had different substances in mind. Panels are ordered by the share choosing a ban. Data: Kieskompas VIP panel, 13–22 October 2026, $n = 1239$, weighted by iterative post-stratification to CBS margins (age, sex, education, migration background, region) and the 2025 Tweede Kamer vote; weights trimmed at the 99th percentile.

A common patterns seems to hold across most drugs and demographics (Figure 3, except for sex which is therefore excluded). Younger, theoretically educated, urban and left-wing respondents converge on regulated options (specialist shops for cannabis, psychedelics and MDMA; medical supply for speed, cocaine and heroin). Older, practically educated, less urban and right-wing respondents differentiate more sharply between drugs, broadly in line with their current legal status: more support for a ban on illicit drugs and, among right-wing voters, more support for free market sale of alcohol and tobacco. Age and education mainly shape support for a ban of currently illegal drugs. The right-left divide most clearly separates market from specialist-shop support for the legal drugs. Urbanization shows this pattern the least.

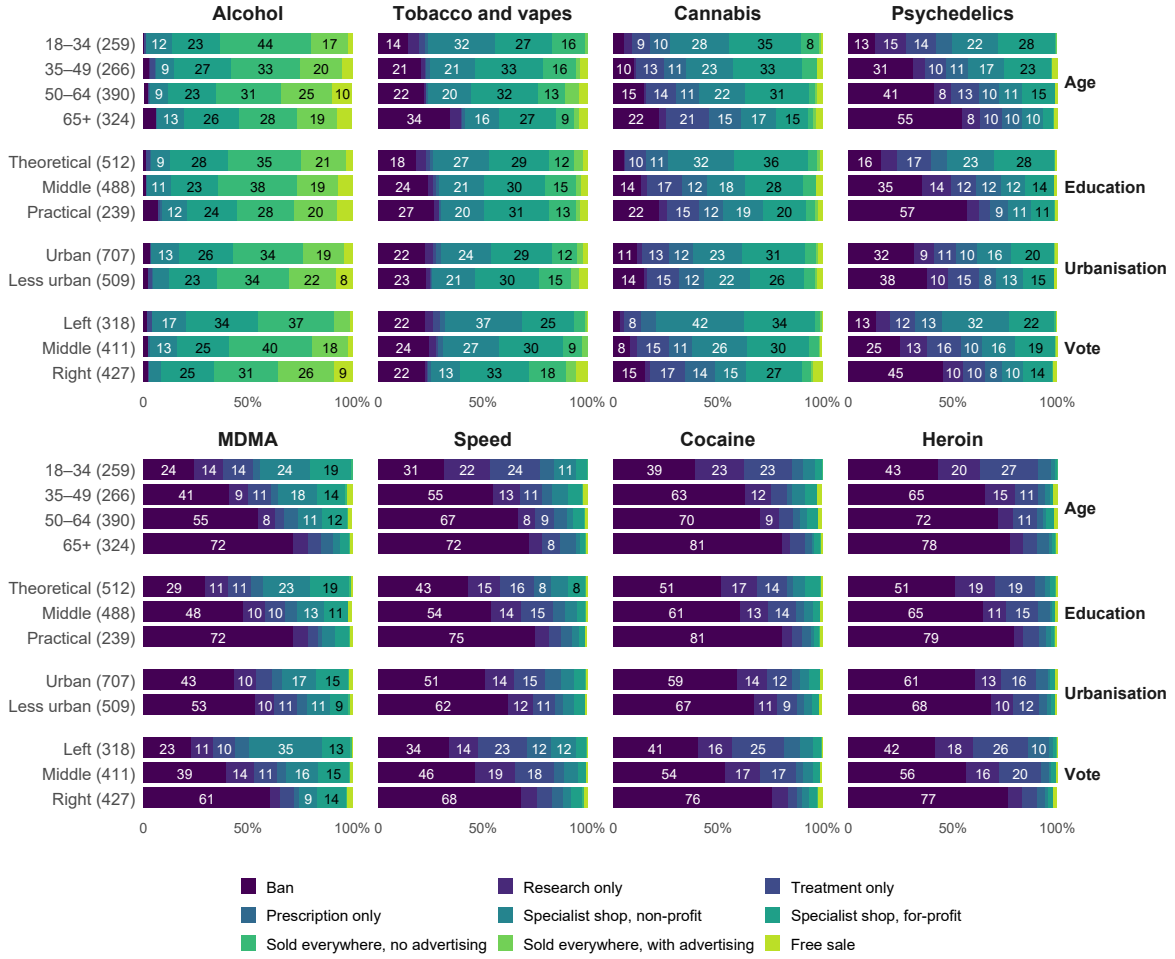
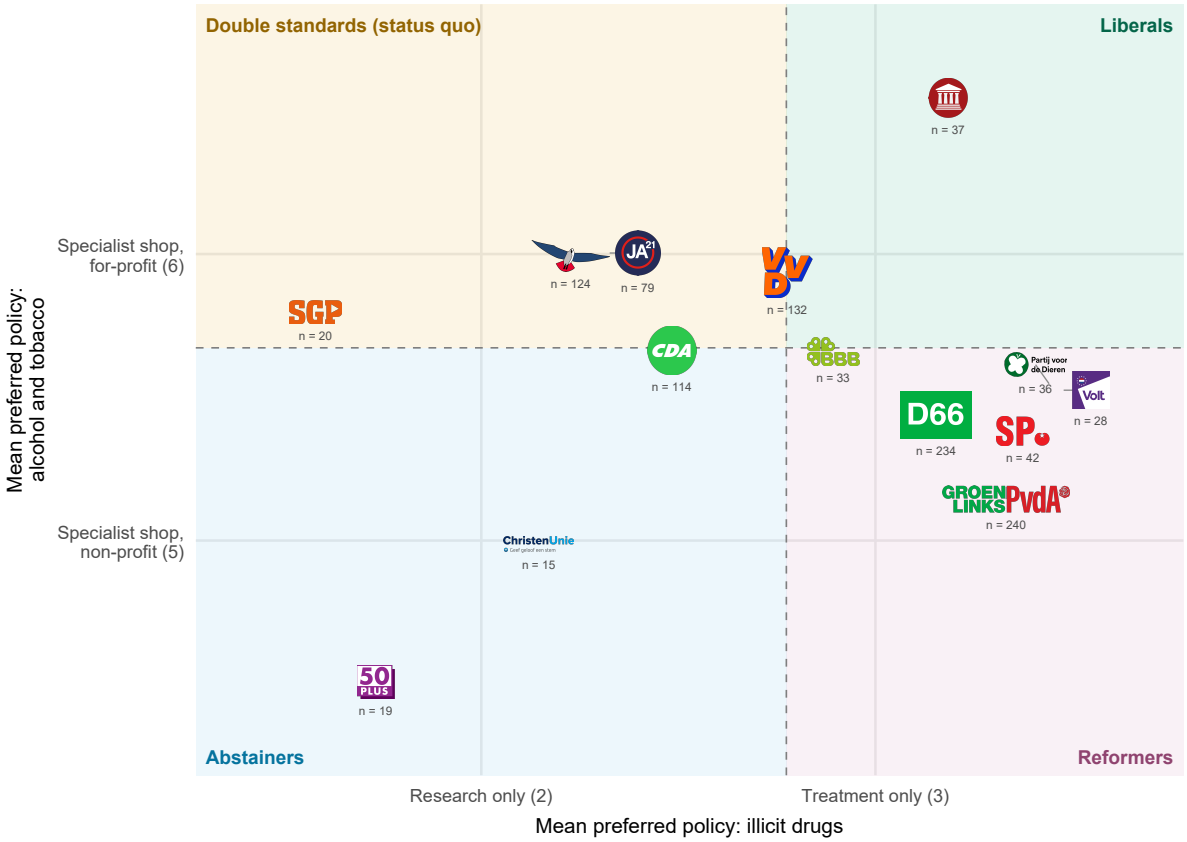


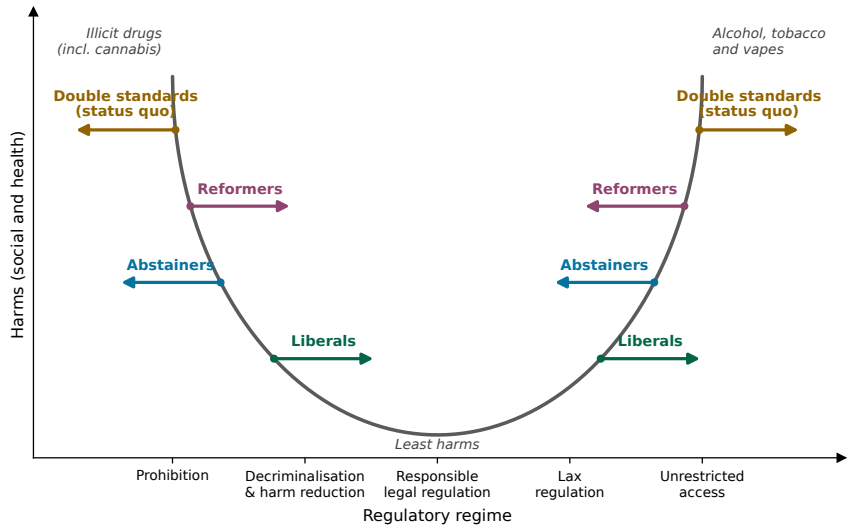
Figure 3. Preferred regulatory model per substance by age, education, urbanisation and vote. Each bar shows the weighted shares of a group choosing each of the nine options, from ban (left) to free sale (right); colours as in Figure 2, labels from 8%. Numbers in brackets are respondents (unweighted). Urban = very high or high address density, less urban = moderate, low or none (23 respondents without urbanisation are not in those rows). Vote (TK 2025) by the party's general left-right position in the Chapel Hill Expert Survey 2024 (Rovny et al., 2025), split into equal thirds of the 0–10 scale: left = SP, PvdD, GroenLinks-PvdA; middle = DENK, Volt, CU, D66, 50PLUS (2019 score), CDA; right = VVD, BBB, SGP, PVV, FvD and JA21 (not rated; placed between VVD and PVV). Other parties, blank and non-voters ($n = 83$) are not in the vote rows.

Figure 4 places the parties by their mean preferred policy for the illicit drugs and for alcohol and tobacco, with the sample means (2.8 and 5.7) splitting the map into four quadrants. PVV and JA21 voters hold the double standards of the status quo: stricter than average on the illicit drugs (2.3, between research and treatment) and more liberal on alcohol and tobacco (6.0, a for-profit specialist shop). Voters of SP, PvdD, GroenLinks-PvdA, Volt and D66 are reformers, the mirror image: more liberal on the illicit drugs (3.2–3.5) and stricter on alcohol and tobacco (5.2–5.6). Only FvD voters are more liberal on both (3.2 and 6.6), while ChristenUnie (2.2 and 5.0) and 50PLUS voters (1.7 and 4.5) are stricter on both. SGP voters are the strictest on the illicit drugs (1.6) but average on alcohol and tobacco (5.8); VVD, CDA and BBB voters sit close to the sample means.

a



b



U-curve adapted from Transform Drug Policy Foundation. Arrows: direction of change each attitude quadrant prefers, left for illicit drugs, right for alcohol, tobacco and vapes.

Figure 4. (a) Parties by mean preferred policy for the illicit drugs (horizontal) and for alcohol and tobacco (vertical), from ban (1) to free sale (9). Per respondent the mean answer over the substances answered, then the weighted mean per party: horizontal over cannabis, psychedelics, MDMA, speed, cocaine and heroin, vertical over alcohol and tobacco. Dashed lines are the sample means and split the map into four quadrants: more liberal than average on the legal drugs only (double standards, the status quo), on both (liberals), on neither (abstainers) or on the illicit drugs only (reformers). Logo size reflects n ($n = 1153$ in total); DENK ($n = 1$), other parties, blank and non-voters are left out. Vote: TK 2025. (b) The four quadrants on the U-curve of harms by regulatory regime, adapted from the Transform Drug Policy Foundation: the arrows show the direction of change each quadrant prefers, for the illicit drugs (left) and for alcohol, tobacco and vapes (right).

5 Conclusion

interpretation demographics One way to think about the demographic divide in the data is as a universalism-particularism divide (Damhuis & Westheuser, 2024): "Liberal – in our parlance: universalist – attitudes have been characterized as marked by moral concerns for the well-being of others (Miles and Vaisey 2015), the critical non-reification of the status quo of cultural practices (Elchardus and Spruyt 2009) and permissiveness (see also Figure 1. Three elements of a cleavage: social structure, group identifications and shared moral outlooks, and their reinforcement by parties' differing ideological supply (our elaboration, based on Bartolini and Mair 1990; Lipset and Rokkan 1967; Bornschier 2010). Broc´ic´ and Miles 2021). [...] Zollinger reconstructs that far right voters' self-descriptions typically center on 'the dignity of a simple, honest life built around selfreliance on hard work and common sense, whereas new left voters highlight their self-perceived openness, tolerance, dynamism, and unorthodox lifestyle' (ibid., 19)"

Importantly, most lay people do not reason in all-encompassing ideological terms. Rather, we consider this divide as providing heuristics for laypeople to reason about complex topics.

interpretation policy preference

Previous measurements of preference for drug regulation have found far higher support for banning of all illicit drugs in our sample. It thus seems that by parsing drug regulation into a spectrum from total ban to total free market, leads to more people considering and subsequently supporting some form of regulation.

6 Discussion

1. new cleavage in Netherlands and europe?
2. Drug regulation requires steady deliberation and incremental steps (MacCoun & Reuter, 2001).

6.1 Limitations and Future Directions

limitations

1. The sample was an opt-in sample from the Kieskompas panel. However, invitations were sent out based on stratification. And the sample was weighted. Nonetheless, the findings may therefore in part reflect sample characteristics. It seems especially likely that the current sample is overrepresented by people with an interest in politics. This can in part be seen in the involvement question: do drugs weight in on your voting behaviour, which a majority gave a response on the top half of the scale. Thus, while our findings clearly suggest that our new measurement leads to more people supporting some form of regulation, replications, preferably using probability-samples, are needed to confirm this.
2. For the illicit drugs we see support for banning between 34 and 64 percent.
3. The attitudes items don't explain currently legal drugs well. In part perhaps because we are missing arguments in favour of commercialisation, beyond competing with the illicit market. In other words, why don't people want to go to government stores for alcohol? Out of laziness? Distrust of the government?

future directions

1. This is a null meting. Future work may explore experimental applications. For example, there is a status-quo bias. People might not be familiar with the status quo. It would interesting to see the effect of simply sharing the status quo with people.

References

Ajzen, I., & Fishbein, M. (1977). Attitude-behavior relations: A theoretical analysis and review of empirical research. *Psychological Bulletin*, *84*(5), 888–918. https://doi.org/10.1037/0033-2909.84.5.888

Bergquist, M., Nilsson, A., Harring, N., & Jagers, S. C. (2022). Meta-analyses of fifteen determinants of public opinion about climate change taxes and laws. *Nature Climate Change*, *12*(3), 235–240. https://doi.org/10.1038/s41558-022-01297-6

Burger, J., et al. (2022). Reporting standards for psychological network analyses in cross-sectional data. *Psychological Methods*.

Chambon, M., Dalege, J., Elberse, J. E., & van Harreveld, F. (2026). Network psychometrics in practice: A practical framework for designing empirical studies that utilize a psychological network approach. *Behavior Research Methods*, *58*, 92.

Citizens' Assembly on Drugs Use. (2024). *Report of the Citizens' Assembly on Drugs Use*. Dublin.

Damhuis, K., & Westheuser, L. (2024). Cleavage politics in ordinary reasoning: How common sense divides. *European Societies*, *26*(4), 1195–1231.

Epskamp, S., Borsboom, D., & Fried, E. I. (2018). Estimating psychological networks and their accuracy. *Behavior Research Methods*, *50*, 195–212.

Felson, J., Adamczyk, A., & Thomas, C. (2019). How and why have attitudes about cannabis legalization changed so much? *Social Science Research*, *78*, 12–27.

Gallup. (2023). *Grassroots support for legalizing marijuana hits record 70%*. Washington, DC.

Greer, A. M., & Ritter, A. (2019). "It's about bloody time": Perceptions of people who use drugs regarding drug law reform. *International Journal of Drug Policy*, *64*, 40–46.

Hamrak, B., & Simonovits, G. (2026). Status quo bias drives public support for drug policy in the USA. *Nature Human Behaviour*, *10*(8), 1487–1494.

Haslbeck, J. M. B., Borsboom, D., & Waldorp, L. J. (2021). Moderated network models. *Multivariate Behavioral Research*, *56*(2), 256–287.

Haslbeck, J. M. B., & Waldorp, L. J. (2020). mgm: Estimating time-varying mixed graphical models in high-dimensional data. *Journal of Statistical Software*, *93*(8), 1–46.

Hoffstadt, M., et al. (2026). ReMoDe. *British Journal of Mathematical and Statistical Psychology*.

MacCoun, R. J., & Reuter, P. (2001). *Drug war heresies*. Cambridge University Press.

Rovny, J., Polk, J., Bakker, R., Hooghe, L., Jolly, S., Marks, G., Steenbergen, M., & Vachudova, M. A. (2025). The 2024 Chapel Hill Expert Survey on political party positioning in Europe: Twenty-five years of party positional data. *Electoral Studies*, *97*, 102981. https://doi.org/10.1016/j.electstud.2025.102981

Saris, W. E., & Gallhofer, I. N. (2004). Operationalization of social science concepts by intuition. *Quality & Quantity*, *38*(3), 235–258. https://doi.org/10.1023/B:QUQU.0000031328.25370.e9

Tobacco and Vapes Act. (2026). Tobacco and Vapes Act 2026 (United Kingdom), c. 12.

van Wingerde, K., Bisschop, L., & Brongers, F. (2023). *Onbedoeld ondermijnen: Verkennend onderzoek naar de wijze waarop de Nederlandse overheid onbedoeld de georganiseerde drugscriminaliteit kan faciliteren*. Erasmus School of Law, in opdracht van het WODC. Rotterdam.

Vysevkova, V., Rabinska, D., Thomas, J. M., Farrow, C., & Reynolds, J. (2026). Explaining support for environmental public policies through the value-belief-norm theory: A systematic review with meta-analysis. *Journal of Environmental Psychology*, *112*, 103071. https://doi.org/10.1016/j.jenvp.2026.103071

Wilkins, C., Tremewan, J., Rychert, M., Atkinson, Q., Fischer, K., & Lavender Forsyth, G. A. (2022). Predictors of voter support for the legalization of recreational cannabis use and supply via a national referendum. *International Journal of Drug Policy*, *99*, 103442.

A Supplementary figures and tables

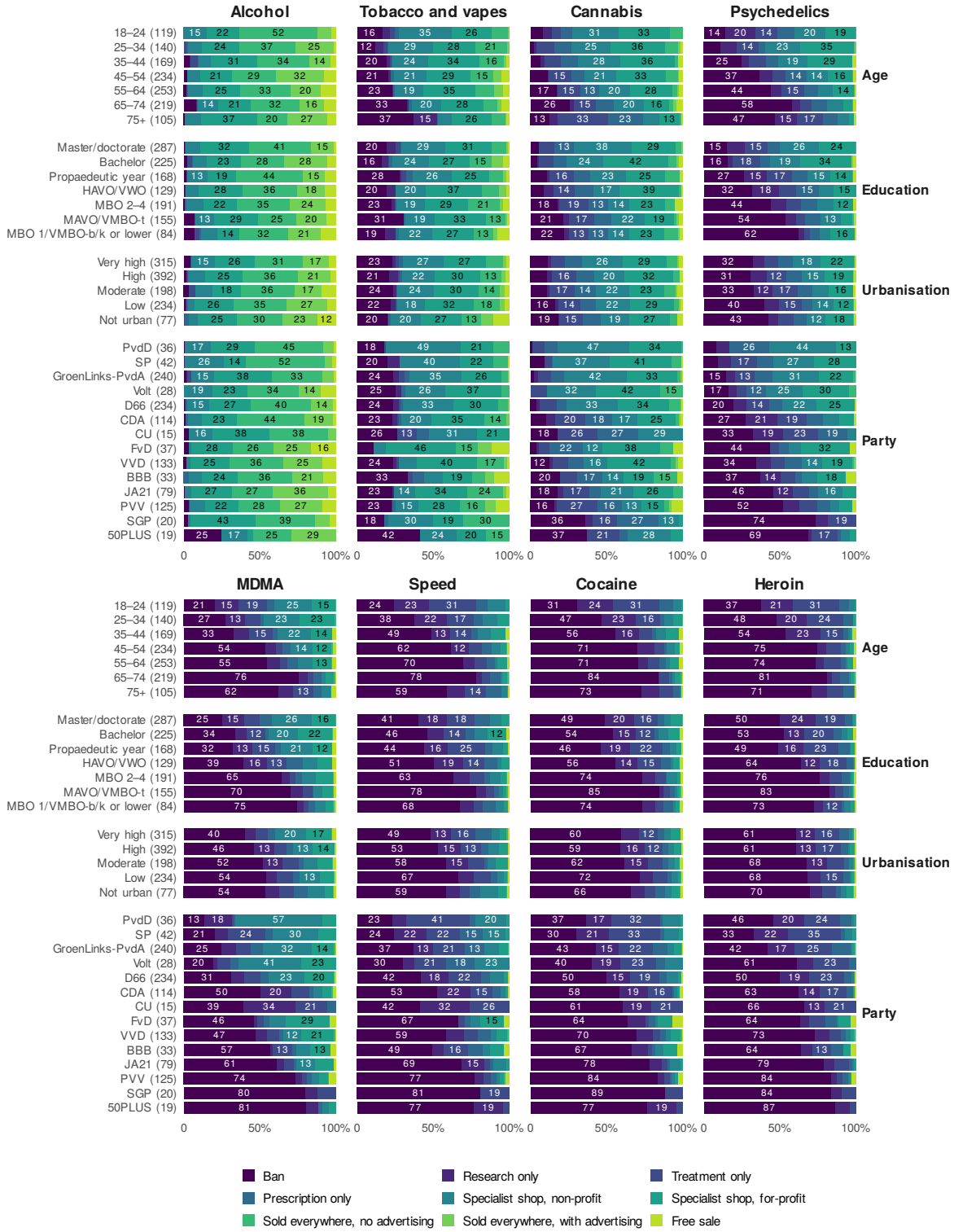


Figure 5. As Figure 3, with detailed categories: age in seven groups, education in seven levels, urbanisation in five levels of address density and vote per party (TK 2025), parties ordered by their share choosing a ban, pooled over the substances; labels from 12%. DENK, other parties, blank and non-voters are left out.

Table 1. Weighted Pearson correlations between the preferred policy for each illicit drug (1 = ban … 9 = free sale) and the attitude items (1 = strongly disagree … 7 = strongly agree), pairwise complete cases; "no opinion" excluded. Items in the survey wording, sorted by the mean absolute correlation over the drugs ($|\bar{r}|$).

Item	Cannabis	Psych.	MDMA	Cocaine	Speed	Heroin	$\lvert\bar{r}\rvert$
Nederland moet zich houden aan het internationale verbod op drugs.	−0.53	−0.57	−0.58	−0.45	−0.47	−0.36	0.49
Drugsgebruik is minder gevaarlijk als de verkoop wordt toegestaan.	0.47	0.50	0.54	0.43	0.44	0.29	0.44
Drugsgebruik is slecht voor de normen en waarden van onze samenleving.	−0.50	−0.52	−0.52	−0.38	−0.43	−0.29	0.44
Het bezit van drugs voor eigen gebruik moet niet strafbaar zijn.	0.49	0.51	0.52	0.41	0.42	0.30	0.44
Als een middel legaal te koop is, raken meer mensen eraan verslaafd.	−0.50	−0.51	−0.47	−0.38	−0.42	−0.30	0.43
Er zou meer politie moeten worden ingezet voor de aanpak van drugs.	−0.40	−0.45	−0.51	−0.37	−0.43	−0.33	0.41
Mensen moeten zelf kunnen beslissen of ze schadelijke middelen gebruiken.	0.50	0.46	0.44	0.39	0.37	0.32	0.41
Een verbod werkt goed om jongeren te beschermen tegen de gevaren van verslavende middelen.	−0.41	−0.44	−0.46	−0.39	−0.41	−0.33	0.41
Er zou in Nederland minder criminaliteit zijn als de verkoop van drugs wordt toegestaan.	0.42	0.44	0.49	0.38	0.38	0.32	0.41
We moeten streven naar een samenleving waarin niemand verslavende middelen gebruikt.	−0.43	−0.40	−0.41	−0.30	−0.34	−0.25	0.36
Mensen die verslaafd zijn zoeken eerder hulp als drugs niet verboden zijn.	0.33	0.37	0.38	0.33	0.34	0.31	0.34
Mensen die softdrugs gebruiken, gaan daarna ook vaak harddrugs gebruiken.	−0.44	−0.40	−0.36	−0.27	−0.27	−0.21	0.32
Drugsgebruikers zijn medeverantwoordelijk voor drugscriminaliteit.	−0.28	−0.34	−0.33	−0.31	−0.30	−0.19	0.29
Het is verkeerd om geld te verdienen aan middelen waar mensen verslaafd aan kunnen raken.	−0.37	−0.31	−0.31	−0.23	−0.22	−0.25	0.28
De verkoop van drugs toestaan zou Nederland veel geld opleveren.	0.34	0.29	0.27	0.21	0.22	0.18	0.25
Als een middel verboden is, gebruiken minder mensen het.	−0.29	−0.26	−0.25	−0.25	−0.23	−0.21	0.25
Veel verboden harddrugs zijn ongezonder dan alcohol en tabak.	−0.20	−0.29	−0.31	−0.25	−0.25	−0.17	0.24
Mensen die verslaafd zijn hebben dat aan zichzelf te danken.	−0.20	−0.23	−0.21	−0.09	−0.19	−0.08	0.17

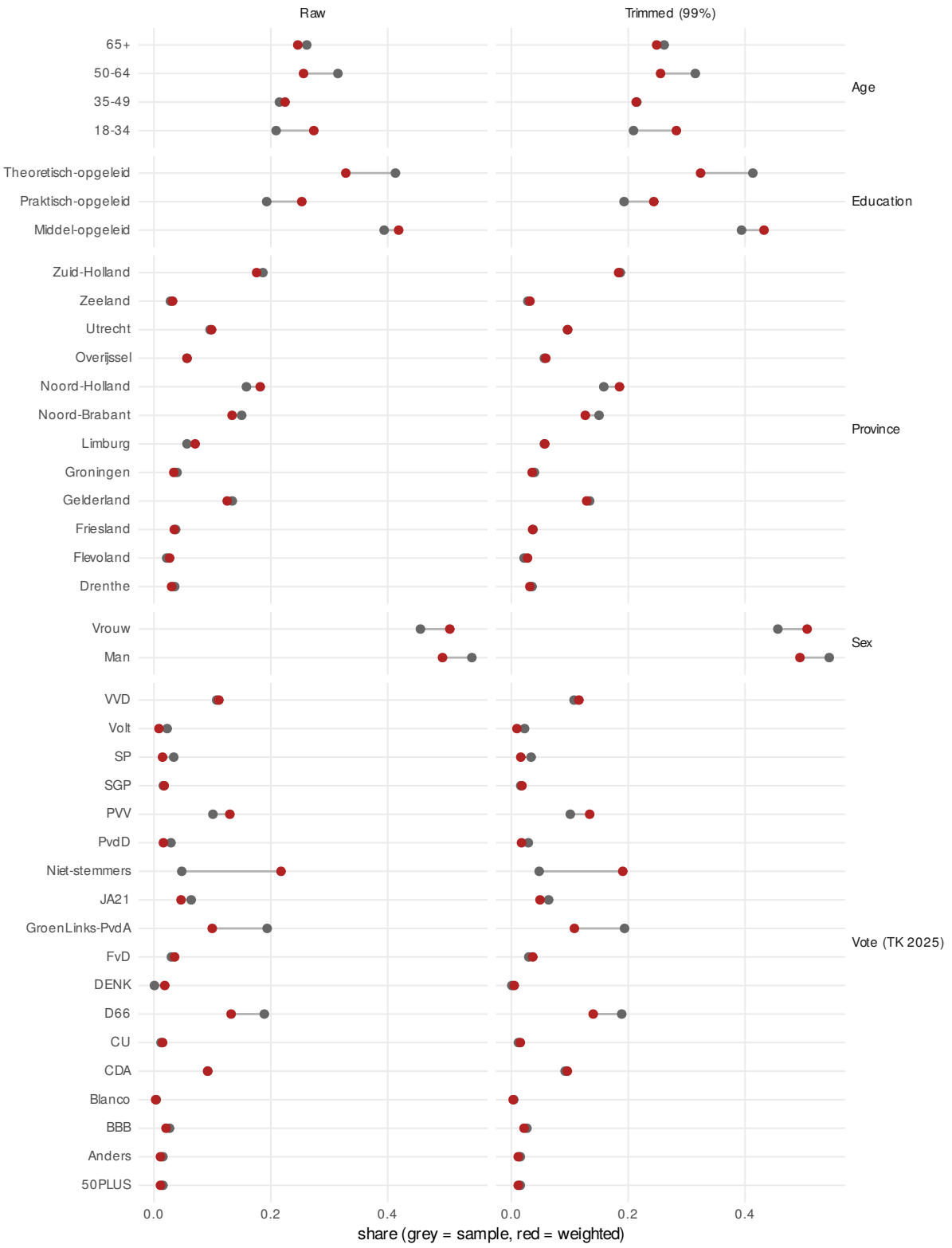


Figure 6. Share of each category of the weighting variables in the sample (grey) and after weighting (red), for the raw weights and the weights trimmed at the 99th percentile (used in all analyses). $n = 1239$.